

RAYMOND AGDM, AB, Canada

WMO: 715360

Lat: 49.487N

Lon: 112.675W

Elev: 3118

StdP: 13.11

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.8	-11.5	-23.7	1.7	-17.5	-17.0	2.4	-11.3	35.9	41.2	31.5	39.1	5.6	230	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	88.0	61.2	84.6	60.5	81.3	59.7	65.3	79.7	63.5	78.5	61.7	76.9	9.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	87.5	71.1	58.0	80.7	69.5	55.8	74.4	66.6	31.9	79.8	30.5	78.6	29.1	77.3	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.2	27.4	23.8	DB	-27.7	94.1	6.3	3.6	-32.2	96.7	-35.9	98.8	-39.4	100.8	-43.9	103.4	
			WB	-27.5	69.4	6.2	2.0	-32.0	70.9	-35.7	72.1	-39.2	73.2	-43.7	74.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	23.9	21.9	32.5	42.0	51.9	58.6	64.8	63.4	55.4	43.6	32.2	22.0	(6)
	DBStd	18.92	15.22	15.59	13.28	9.17	8.01	5.99	5.41	6.18	8.22	9.82	13.49	14.80	(7)
	HDD50	4169	808	785	545	264	74	5	0	1	39	241	540	867	(8)
	HDD65	8255	1273	1205	1007	691	413	208	70	105	301	665	985	1332	(9)
	CDD50	1542	0	0	3	23	131	263	460	416	200	41	5	0	(10)
	CDD65	152	0	0	0	0	5	16	65	54	12	0	0	0	(11)
	CDH74	3114	0	0	0	17	166	288	1184	1076	372	11	0	0	(12)
	CDH80	976	0	0	0	1	32	58	399	375	111	0	0	0	(13)
(14) Wind	WSAvg	10.2	11.9	10.3	11.0	11.3	10.2	9.8	8.0	7.9	9.2	10.5	11.3	11.1	(14)
(15) Precipitation	PrecAvg	14.9	0.5	0.4	0.7	1.2	2.2	3.5	1.6	1.4	1.3	0.9	0.6	0.6	(15)
	PrecMax	22.8	0.8	0.9	1.6	2.4	4.2	8.9	5.5	3.3	3.7	2.7	1.3	1.1	(16)
	PrecMin	8.0	0.1	0.0	0.2	0.5	0.6	0.6	0.1	0.0	0.1	0.2	0.2	0.0	(17)
	PrecStd	3.5	0.2	0.2	0.4	0.5	1.0	2.0	1.2	0.8	0.9	0.6	0.3	0.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	56.4	64.3	76.4	83.7	86.7	92.9	91.8	88.2	75.0	65.5	53.0	(19)
		MCWB	42.8	42.3	46.3	51.8	56.6	64.0	63.6	60.9	58.4	52.8	48.5	41.7	(20)
	2%	DB	48.9	50.0	58.8	68.6	78.9	80.7	87.9	88.0	83.3	69.2	57.8	47.8	(21)
		MCWB	40.4	39.3	43.9	48.3	55.3	60.3	63.5	60.0	57.0	51.1	45.1	39.6	(22)
	5%	DB	44.7	45.7	54.6	63.1	73.8	77.3	84.4	84.3	78.3	63.6	53.5	44.1	(23)
		MCWB	37.5	36.9	42.2	46.0	53.4	59.1	62.6	59.5	55.5	48.8	42.8	36.6	(24)
	10%	DB	41.8	41.9	50.3	58.4	68.9	73.1	80.8	80.3	72.6	58.6	49.1	40.9	(25)
		MCWB	35.7	34.8	39.9	43.9	51.3	57.1	61.8	58.9	53.8	46.4	40.4	34.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	43.7	47.7	52.8	58.1	66.6	69.3	65.0	61.2	55.4	50.4	43.1	(27)
		MCDB	53.5	54.1	60.7	72.3	78.4	79.9	82.3	79.8	79.3	70.3	62.7	51.3	(28)
	2%	WB	40.3	39.7	45.2	49.2	56.1	63.2	66.4	63.3	58.6	52.1	46.4	39.6	(29)
		MCDB	47.7	48.9	56.9	66.6	75.2	77.3	80.5	80.0	77.2	66.3	57.0	46.9	(30)
	5%	WB	38.0	37.3	42.7	46.7	54.1	60.5	64.7	61.6	56.7	49.5	43.1	36.9	(31)
		MCDB	44.1	44.9	54.0	61.7	71.3	73.4	79.1	78.6	75.8	61.8	52.0	43.2	(32)
	10%	WB	35.9	35.1	40.1	44.4	52.4	58.4	63.0	60.1	54.5	46.9	40.5	34.7	(33)
		MCDB	41.2	41.5	49.4	57.1	66.9	70.7	77.8	76.7	71.5	57.6	48.6	40.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.9	20.3	22.3	23.5	22.6	27.9	28.1	25.4	22.1	19.6	18.3	(35)
		MCDBR	18.5	22.7	26.2	31.1	32.2	29.9	34.7	36.4	35.9	28.8	24.9	18.7	(36)
		MCWBR	12.2	14.0	14.7	15.0	13.5	13.6	14.7	13.8	14.6	14.4	14.6	12.8	(37)
	5% WB	MCDBR	18.3	22.4	24.9	29.1	29.0	26.3	29.8	31.4	33.4	27.5	23.6	19.0	(38)
		MCWBR	12.5	14.7	14.5	14.7	12.7	13.3	14.4	13.5	14.3	14.3	14.4	13.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.285	0.322	0.334	0.331	0.333	0.352	0.314	0.281	0.258	0.236	(40)
	taud		2.453	2.482	2.476	2.383	2.401	2.454	2.460	2.383	2.463	2.534	2.501	2.454	(41)
	Ebn at Noon		264	289	297	294	293	294	292	280	282	276	259	256	(42)
	Edh at Noon		19	23	28	35	36	34	33	34	29	22	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg		394	713	1109	1536	1829	2042	2250	1787	1270	792	444	315	(44)
	RadStd		33	53	95	103	158	152	118	109	104	79	34	31	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (61 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47) N/A

Nomenclature: See separate page